```
    _    ____  _____  __   _____  _   _
   / \  |  _ \| ____| \ \ / / _ \| | | |
  / _ \ | |_) |  _|    \ V / | | | | | |
 / ___ \|  _ <| |___    | || |_| | |_| |
/_/   \_\_| \_\_____|   |_| \___/ \___/

 ____    _    ____  _____ ____ ___
| __ )  / \  / ___|| ____|  _ \__ \
|  _ \ / _ \ \___ \|  _| | | | |/ /
| |_) / ___ \ ___) | |___| |_| |_|
|____/_/   \_\____/|_____|____/(_)
```

echo $(whoami)

- Principle Software Engineer
- Co-founder and Chief Terminal Nerd @ PHL Code Club 🖥

Here is me with me cat, Fritzy =3

Setting Expectations

What this is:

- An overview of various database paradigms and their uses
- A look at some production headaches you might have with your database
- Some time for me to yap about technology :D

What this isn't:

- An exhaustive list of databases
- A full deep dive into database architectures and subsystems (But I'm happy to come back and talk about that!)

Overview

Topics we will cover:

- What is a database?
- Some key database paradigms, with strengths and weaknesses
  - Key-Value
  - Relational
  - Document
  - Fulltext Search
- Exploring some databases inside Docker

What Is a Database?

Who can tell me what a database is? What are the used for?

It's an organized set of data usually stored on disk. This data is often managed by a Database Management System or DBMS. Generally speaking the data is stored in a manner that makes retrieval and manipulation as efficient as possible.

Database Paradigms

What do I mean when I say a "Database Paradigm"?

Data can take many shapes. It might be a social network, a blog, or telemetry data. We want to store and query our data in a way that matches the shape of the data.

Key-Value Stores

Can anyone give me an example of a Key-Value Store?

At their core, Key-Value Stores are no different than HashMaps or Dictionaries. They map a key to a value, and nothing more. This makes lookups **super** fast, but the lookup constraint isn't useful on complex data.

You generally see these used as in memory application caches, such as Redis or Memcached. However, they can also be used for very performant simple sorage such as etcd which powers Kubernetes.

Relational Databases

Can anyone give me an example of a Relational Database?

Relational Databases are defined by structured sets of rows and columns called tables. These tables have strict data schemas that tell you what data types they can hold. They also allow for creating relationships between columns of data which allow you to join tables together efficiently.

Common Relational Databases are PostgreSQL, MySQL, and SQLite. You may also come across "Enterprise" databases such as OracleDB and Microsoft SQL Server.

Document Databases

Can anyone give me an example of a Document Database?

Document Databases are heirarchical sets of key-value pairs, similar to a key-value store, but they allow for nesting documents and support more complex datatypes.

One big feature of most Document Databases is that they schemaless, meaning the data can be any shape when you store it.

Examples of Document Databases include MongoDB and CouchDB.

Full Text Search Engines

Can anyone give me an example of a Full Text Search Engine?

Fulltext Search Engines ingest data from other sources and index it in such a way that makes searching text (hence the name) and ranking results fast and efficient.

These are often secondary to a primary database, such as a Relational or Document DB.

Some examples of Fulltext Search Engines are ElasticSearch, OpenSearch, and Typesense.

Let's Dig In

You can find a git repo with a small docker compose setup to play with here:

repo (https://github.com/gv14982/launchpad-db)

echo $(which gvasquez)

My website! (https://gvasquez.dev/)

PHL Code Club (https://phlcode.club/)